

KIEL HOLTENAU, Germany

WMO: 100460

Lat: 54.3761N

Lon: 10.1434E

Elev: 30

StdP: 100.97

Time Zone: 1.00 (EUC)

Period: 95-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.0	-10.8	1.5	-6.8	-8.8	1.8	-4.8	13.4	5.5	12.2	6.5	2.8	270	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	27.2	19.1	25.1	18.3	23.2	17.5	20.2	25.6	19.1	23.9	18.2	22.5	3.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	22.3	17.2	12.3	21.4	16.2	11.6	20.4	57.8	25.8	54.3	24.0	51.5	22.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.9	7.9	-10.9	31.4	2.5	1.7	-12.7	32.7	-14.1	33.7	-15.5	34.6	-17.3	35.9		
(5)			-11.1	22.2	2.4	0.8	-12.8	22.8	-14.2	23.2	-15.6	23.7	-17.3	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	1.7	2.1	4.5	8.3	11.9	15.0	17.5	17.7	14.5	9.9	5.8	2.2
	DBStd	6.70	4.10	4.06	3.62	3.39	3.04	3.00	2.88	2.75	2.83	3.15	3.41	3.99
	HDD10.0	1152	258	223	173	72	15	1	0	0	1	40	130	241
	HDD18.3	3364	516	456	430	301	200	109	52	46	120	260	376	499
	CDD10.0	896	0	1	2	21	74	149	233	239	136	38	4	0
	CDD18.3	66	0	0	0	0	1	8	26	26	5	0	0	0
	CDH23.3	436	0	0	0	3	14	65	178	153	23	0	0	0
	CDH26.7	86	0	0	0	0	3	16	36	30	2	0	0	0
Wind	WSAvg	3.9	4.4	4.6	4.4	3.7	3.7	3.8	3.6	3.3	3.5	3.8	3.9	4.2
Precipitation	PrecAvg	779	56	45	43	56	52	64	84	82	65	64	93	76
	PrecMax	947	117	74	70	126	78	106	176	210	99	101	203	160
	PrecMin	549	13	12	10	12	17	30	29	35	15	10	34	9
	PrecStd	127	30	20	18	30	20	22	40	53	21	28	49	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.1	16.1	22.2	25.0	29.0	30.1	29.2	26.1	19.2	14.2	11.2
		MCWB	9.1	9.0	10.3	14.4	16.8	20.3	19.9	19.8	18.0	15.2	12.0	9.5
	2%	DB	9.8	10.1	13.1	18.2	21.4	25.0	27.1	27.1	23.2	17.1	12.9	9.9
		MCWB	8.3	8.2	9.1	12.0	14.7	18.0	19.1	19.3	17.1	14.2	11.0	8.8
	5%	DB	8.1	8.8	11.1	16.1	19.2	22.6	25.1	24.9	21.1	15.8	11.2	8.2
		MCWB	7.0	7.6	8.2	11.2	13.7	16.8	18.6	18.6	16.3	13.3	9.5	7.3
	10%	DB	7.1	7.2	9.8	14.1	17.9	20.2	23.2	23.0	19.1	14.2	10.2	7.2
		MCWB	6.0	6.2	7.5	9.9	12.9	15.5	17.6	17.7	15.4	12.2	8.9	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.9	11.3	14.8	17.5	20.9	21.4	21.4	19.3	15.9	12.8	10.4
		MCDB	10.8	11.1	14.1	20.5	23.3	27.1	26.6	26.8	23.9	18.7	14.1	11.4
	2%	WB	8.2	8.4	9.9	13.0	15.7	18.8	20.3	20.0	17.8	14.4	11.2	8.8
		MCDB	9.2	9.5	12.4	17.4	19.9	24.0	25.7	25.3	21.9	16.6	12.8	9.7
	5%	WB	7.0	7.4	8.7	11.6	14.5	17.3	19.2	19.0	16.7	13.5	10.1	7.5
		MCDB	8.0	8.4	10.9	15.6	18.3	21.2	24.0	23.5	20.5	15.4	11.2	8.3
	10%	WB	5.9	6.2	7.4	10.4	13.4	16.1	18.0	18.3	15.7	12.6	9.1	6.4
		MCDB	6.8	7.2	9.1	13.4	16.9	19.7	22.3	22.3	18.8	14.4	10.2	7.1
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	6.3	8.1	8.9	8.7	8.4	8.2	7.4	6.0	4.3	3.8
		MCDBR	4.5	5.0	8.8	11.6	11.8	11.6	11.5	10.9	10.1	7.3	4.7	4.2
		MCWBR	4.1	4.1	6.1	7.0	7.0	6.5	5.7	5.7	5.6	5.1	4.0	4.1
	5% WB	MCDBR	4.4	4.7	8.1	10.6	10.5	10.5	10.8	9.5	9.0	6.5	4.6	4.2
		MCWBR	4.1	4.1	6.0	6.7	6.8	6.3	5.9	5.5	5.5	4.9	4.2	4.2
Clear-Sky Solar Irradiance	taub	0.318	0.339	0.367	0.406	0.398	0.395	0.413	0.410	0.384	0.358	0.335	0.308	
	taud	2.301	2.319	2.269	2.209	2.306	2.343	2.308	2.329	2.383	2.418	2.361	2.295	
	Ebn at Noon	621	742	800	816	848	854	828	806	779	714	598	549	
	Edh at Noon	58	79	103	125	120	117	119	110	92	72	55	48	
All-Sky Solar Radiation	RadAvg	0.50	1.14	2.42	4.10	5.22	5.48	5.14	4.16	3.00	1.59	0.65	0.37	
	RadStd	0.07	0.18	0.23	0.49	0.56	0.40	0.61	0.30	0.31	0.20	0.08	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-149	N/A	N/A

Nomenclature: See separate page